

VLSI Design

Chapter 4&9: Delay

Content

- Normalized Delay
- Linear Delay Model
- Delay of Multistage Logic Networks
 - Compound Gates
 - NAND/NOR Implementations

Normalized Delay $\rightarrow$ Inverter

- Express delay in a process-independent form
- Circuits can be compared based on topology rather than speed of the manufacturing process
- Observe that the delay of an ideal fanout-of-1 inverter with no parasitic capacitance is $\tau = 3RC$
- Normalized Delay (d) relative to this inverter delay:

$$d = \frac{t_{pd}}{\tau}$$

- Hence, the **delay of a fanout-of- h inverter** can be written in normalized form as, (diffusion capacitance = gate capacitance)

$$d = h + 1$$

Linear Delay Model

- RC model showed that delay is a linear function of fanout of a gate
- Designers simplify delay analysis by characterizing a gate by the slope and y-intercept of the function
- In LDM, **normalized delay** is expressed in terms of two components of delay:

$$d = f + p$$

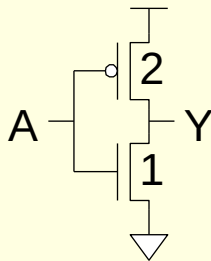
1. The **parasitic delay (p)** is the time for a gate to drive its own internal diffusion capacitance (no load)
2. The **effort delay/stage effort (f)** depends on h and g:

$$f = gh$$

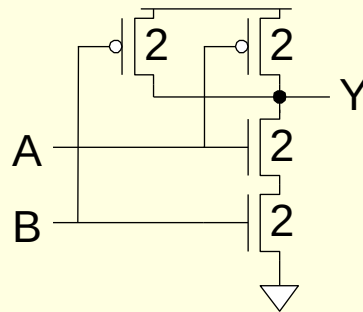
- h (fanout/electrical effort) is the ratio of the capacitance of the external load to input capacitance of the gate
- Complexity (logical effort, g) of the gate. Complex gates have greater logical efforts, indicating that they take longer to drive a given function

Computing Logical Effort

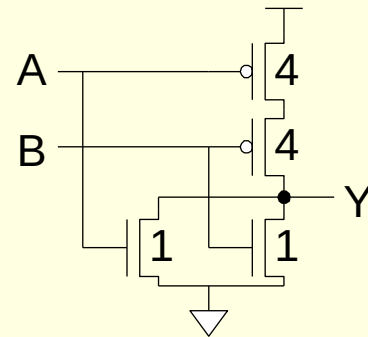
- DEF: *Logical effort is the ratio of the input capacitance of a gate to the input capacitance of an inverter delivering the same output current.*
- Measure from delay vs. fanout plots
- Or estimate by counting transistor widths



$$C_{in} = 3$$
$$g = 3/3$$



$$C_{in} = 4$$
$$g = 4/3$$



$$C_{in} = 5$$
$$g = 5/3$$

Catalog of Gates

■ Logical effort of common gates

Gate type	Number of inputs				
	1	2	3	4	n
Inverter	1				
NAND		4/3	5/3	6/3	$(n+2)/3$
NOR		5/3	7/3	9/3	$(2n+1)/3$
Tristate / mux	2	2	2	2	2
XOR, XNOR		4, 4	6, 12, 6	8, 16, 16, 8	

Parasitic Delay

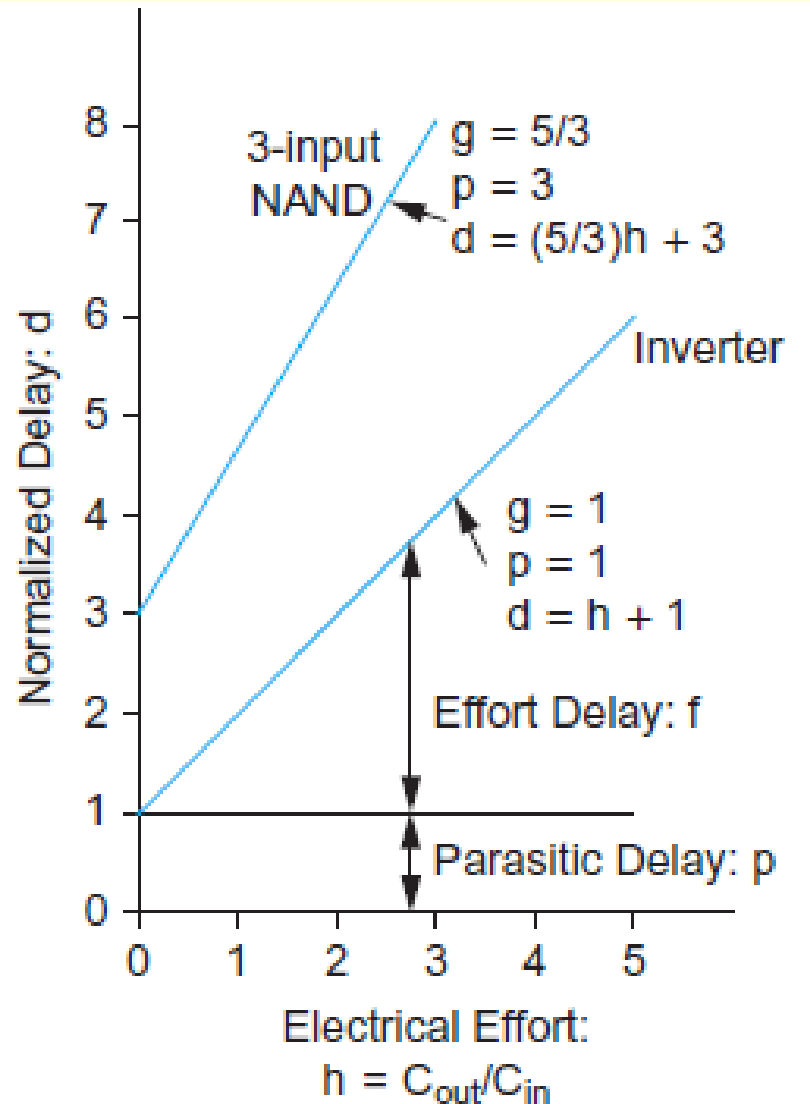
- The parasitic delay of a gate is the delay of the gate when it drives zero load. It can be estimated with RC delay models.
- The inverter has three units of diffusion capacitance on the output, so the parasitic delay is $3RC = \tau$
- The 3-input NAND and NOR each have 9 units of diffusion capacitance on the output, so the parasitic delay is three times as great.
- A crude method for hand calculations is to count only diffusion capacitance on the output node.

Catalog of Gates

- Parasitic delay of common gates
 - In multiples of p_{inv} (≈ 1)

Gate type	Number of inputs				
	1	2	3	4	n
Inverter	1				
NAND		2	3	4	n
NOR		2	3	4	n
Tristate / mux	2	4	6	8	2n
XOR, XNOR		4	6	8	

Linear Delay Model...continue



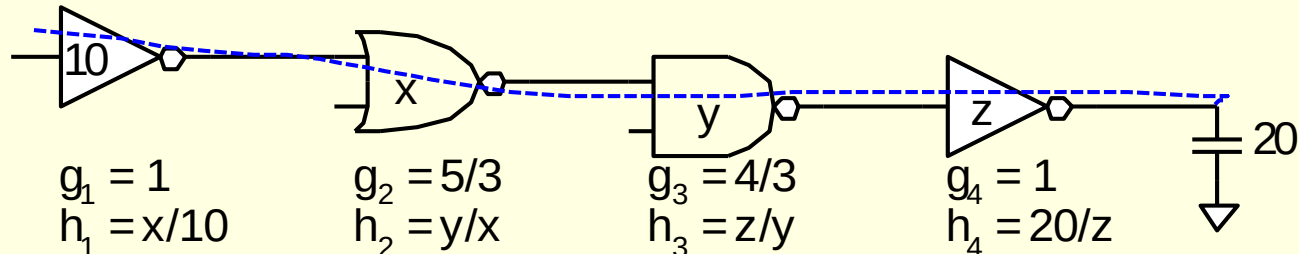
Multistage Logic Networks

- Logical effort generalizes to multistage networks

- *Path Logical Effort* $G = \prod g_i$

- *Path Electrical Effort* $H = \frac{C_{\text{out-path}}}{C_{\text{in-path}}}$

- *Path Effort* $F = \prod f_i = \prod g_i h_i$



Multistage Logic Networks

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- *Path Effort* $F = \prod f_i = \prod g_i h_i$

- Can we write $F = GH$?

Paths that Branch

- No! Consider paths that branch:

$G =$

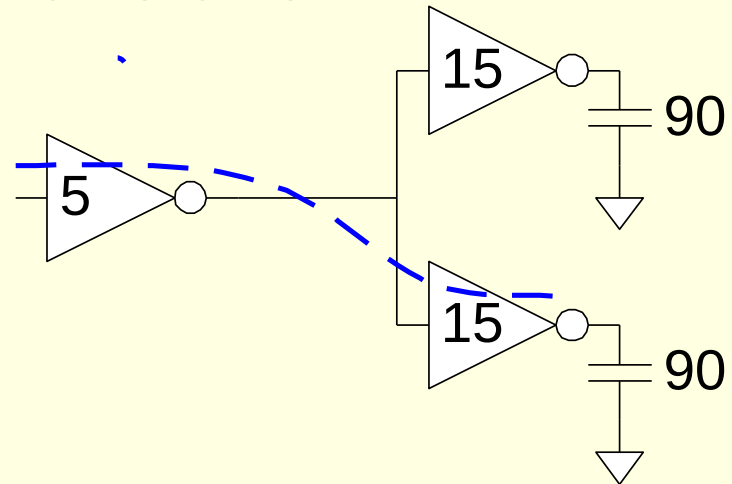
$H =$

$GH =$

$h_1 =$

$h_2 =$

$F = GH?$



Paths that Branch

- No! Consider paths that branch:

$$G = 1$$

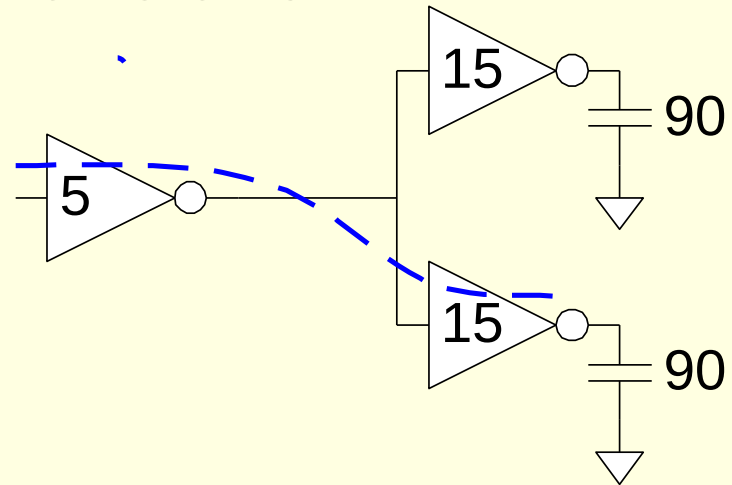
$$H = 90 / 5 = 18$$

$$GH = 18$$

$$h_1 = (15 + 15) / 5 = 6$$

$$h_2 = 90 / 15 = 6$$

$$F = g_1 g_2 h_1 h_2 = 36 = 2GH$$



Branching Effort

- Introduce *branching effort*
 - Accounts for branching between stages in path

$$b = \frac{C_{\text{on path}} + C_{\text{off path}}}{C_{\text{on path}}}$$

- Path branching effort B

$$B = \prod b_i$$

- Now we compute the path effort
 - $F = GBH$

Multistage Delays

- Path Effort Delay $D_F = \sum f_i$
- Path Parasitic Delay $P = \sum p_i$
- Path Delay $D = \sum d_i = D_F + P$

Designing Fast Circuits

$$D = \sum d_i = D_F + P$$

- Delay is smallest when each stage bears same effort

$$\hat{f} = g_i h_i = F^{\frac{1}{N}}$$

- Thus minimum delay of N stage path is

$$D = NF^{\frac{1}{N}} + P$$

- This is a **key** result of logical effort
 - Find fastest possible delay
 - Doesn't require calculating gate sizes

Gate Sizes

- How wide should the gates be for least delay?

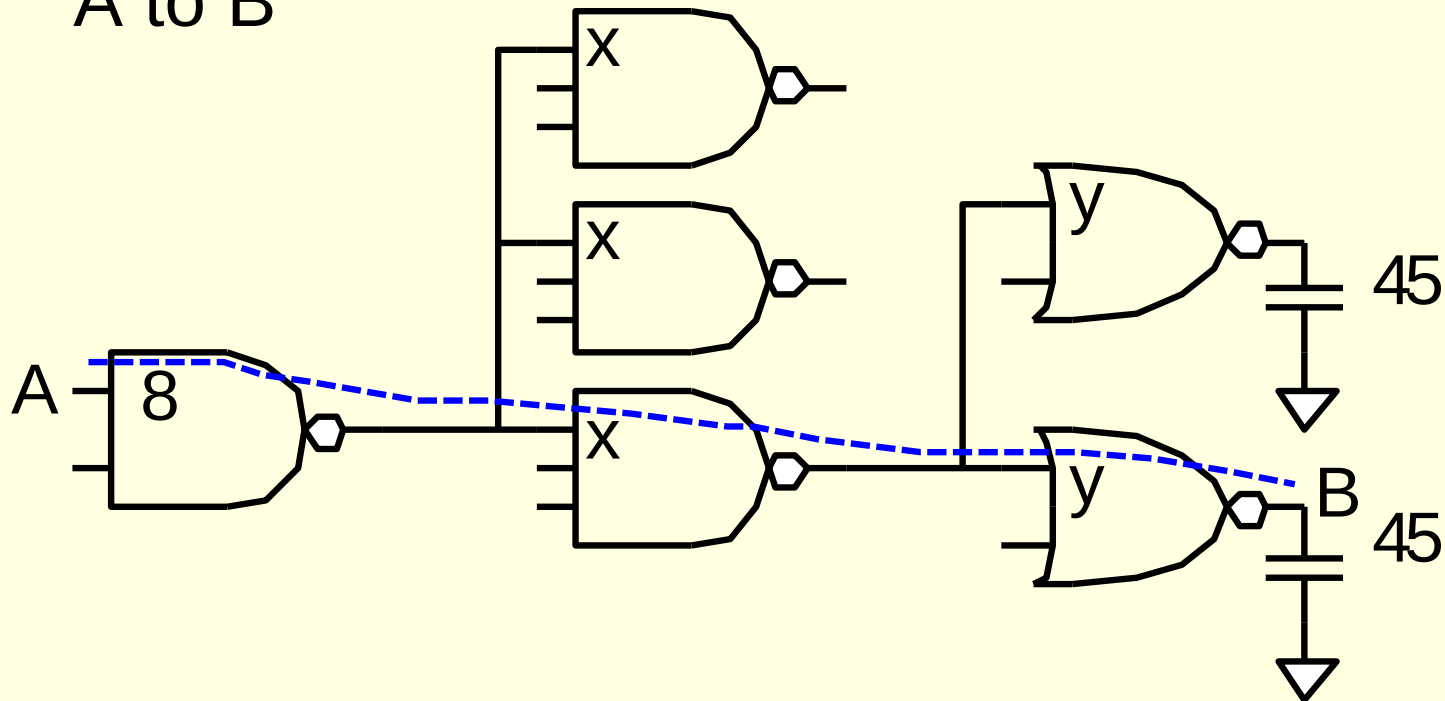
$$\hat{f} = gh = g \frac{C_{out}}{C_{in}}$$

$$\Rightarrow C_{in_i} = \frac{g_i C_{out_i}}{\hat{f}}$$

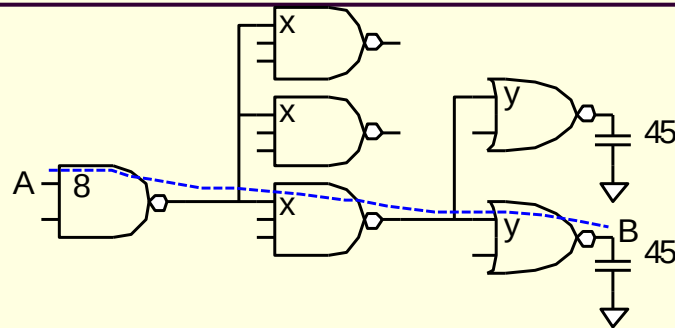
- Working backward, apply capacitance transformation to find input capacitance of each gate given load it drives.
- Check work by verifying input cap spec is met.

Example: 3-stage path

- Select gate sizes x and y for least delay from A to B



Example: 3-stage path



Logical Effort $G =$

Electrical Effort $H =$

Branching Effort $B =$

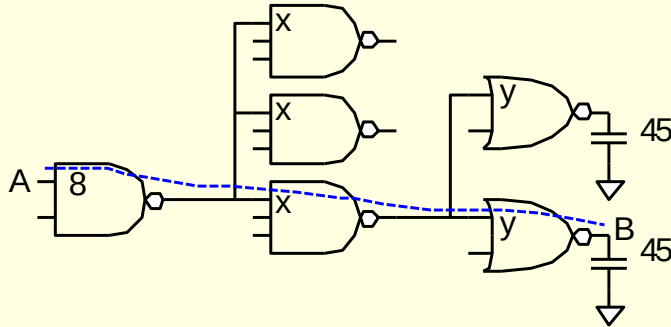
Path Effort $F =$

Best Stage Effort $\hat{f} =$

Parasitic Delay $P =$

Delay $D =$

Example: 3-stage path



Logical Effort

$$G = (4/3) * (5/3) * (5/3) = 100/27$$

Electrical Effort

$$H = 45/8$$

Branching Effort

$$B = 3 * 2 = 6$$

Path Effort

$$F = GBH = 125$$

Best Stage Effort

$$\hat{f} = \sqrt[3]{F} = 5$$

Parasitic Delay

$$P = 2 + 3 + 2 = 7$$

Delay

$$D = 3 * 5 + 7 = 22 = 4.4 FO4$$

Example: 3-stage path

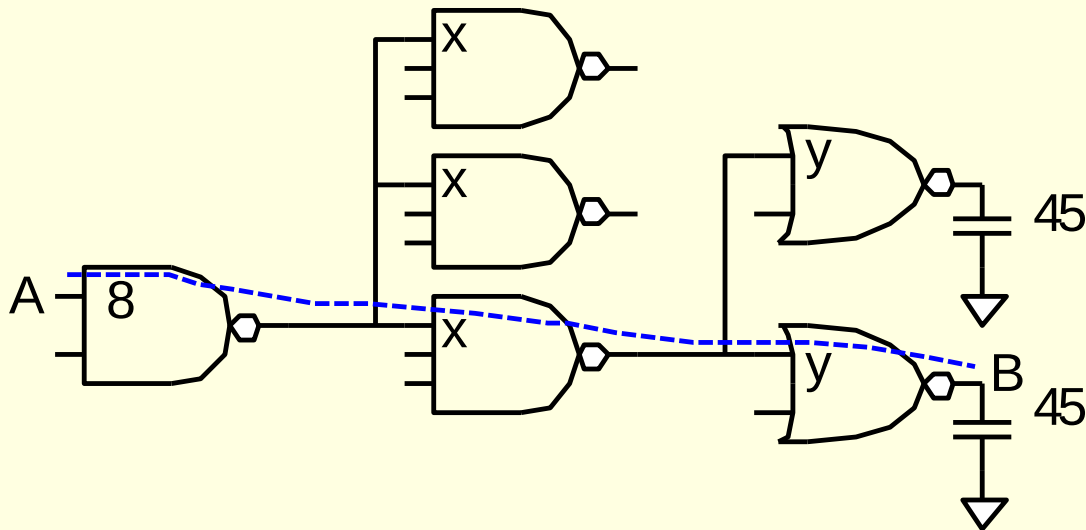
- Work backward for sizes

$y =$

$x =$

$$\hat{f} = gh = g \frac{C_{out}}{C_{in}}$$

$$\Rightarrow C_{in_i} = \frac{g_i C_{out_i}}{\hat{f}}$$



Example: 3-stage path

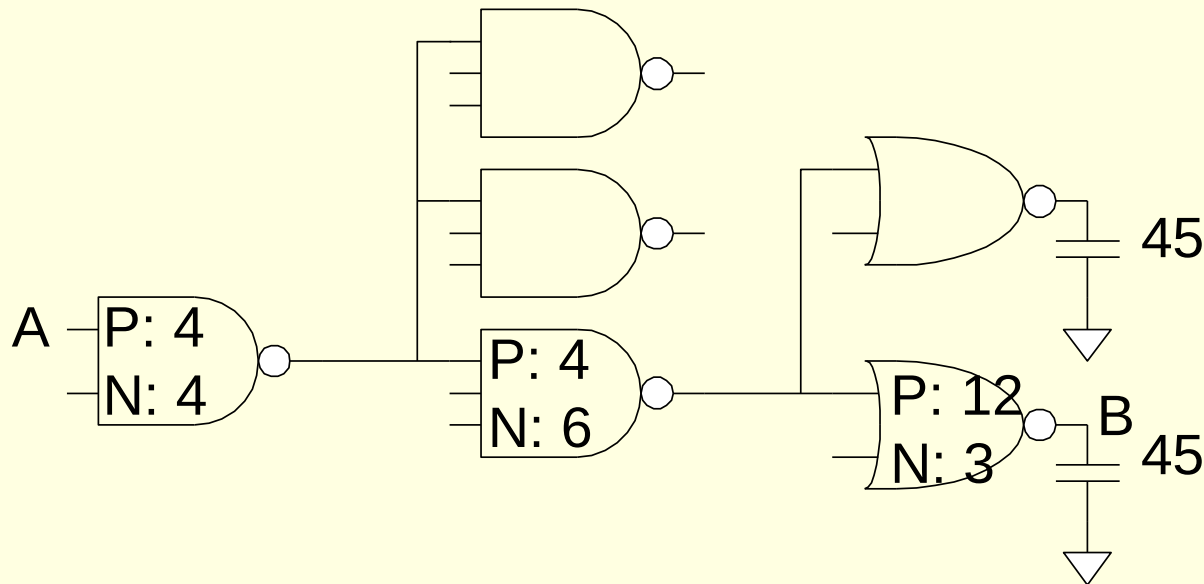
- Work backward for sizes

$$y = 45 * (5/3) / 5 = 15$$

$$x = (15 * 2) * (5/3) / 5 = 10$$

$$\hat{f} = gh = g \frac{C_{out}}{C_{in}}$$

$$\Rightarrow C_{in_i} = \frac{g_i C_{out_i}}{\hat{f}}$$

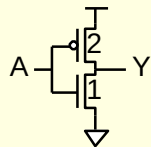
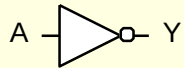


Compound Gates

■ Logical Effort of compound gates

Unit Inverter

$$Y = \overline{A}$$

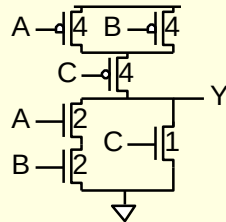
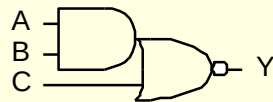


$$g_A = 3/3$$

$$p = 3/3$$

AOI21

$$Y = A \cdot B + C$$



$$g_A = 6/3$$

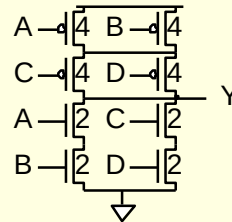
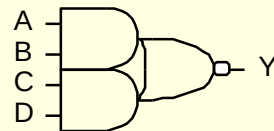
$$g_B = 6/3$$

$$g_C = 5/3$$

$$p = 7/3$$

AOI22

$$Y = A \cdot B + C \cdot D$$



$$g_A =$$

$$g_B =$$

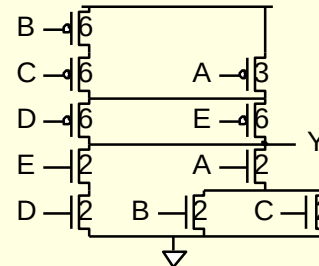
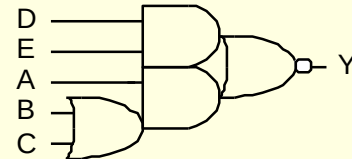
$$g_C =$$

$$g_D =$$

$$p =$$

Complex AOI

$$Y = A \cdot (B + C) + D \cdot E$$



$$g_A =$$

$$g_B =$$

$$g_C =$$

$$g_D =$$

$$g_E =$$

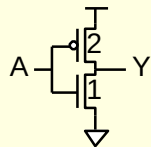
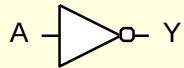
$$p =$$

Compound Gates

■ Logical Effort of compound gates

Unit Inverter

$$Y = \overline{A}$$

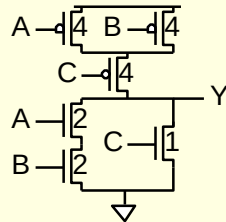
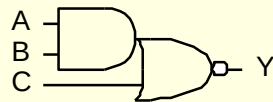


$$g_A = 3/3$$

$$p = 3/3$$

AOI21

$$Y = A \cdot B + C$$



$$g_A = 6/3$$

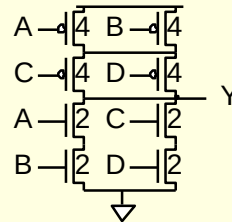
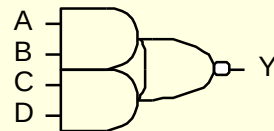
$$g_B = 6/3$$

$$g_C = 5/3$$

$$p = 7/3$$

AOI22

$$Y = A \cdot B + C \cdot D$$



$$g_A = 6/3$$

$$g_B = 6/3$$

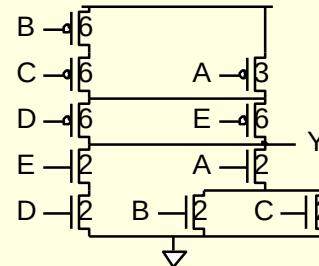
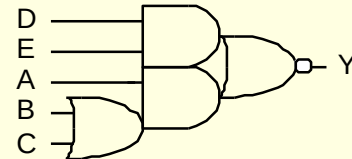
$$g_C = 6/3$$

$$g_D = 6/3$$

$$p = 12/3$$

Complex AOI

$$Y = A \cdot (B + C) + D \cdot E$$



$$g_A = 5/3$$

$$g_B = 8/3$$

$$g_C = 8/3$$

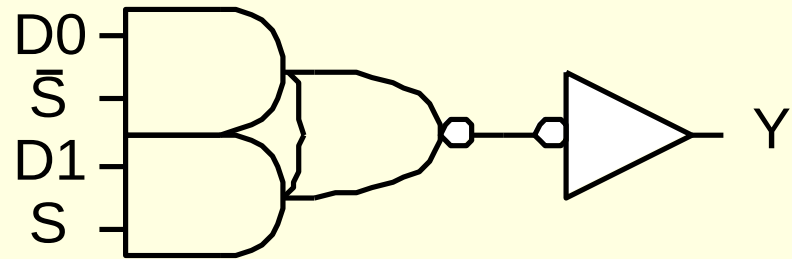
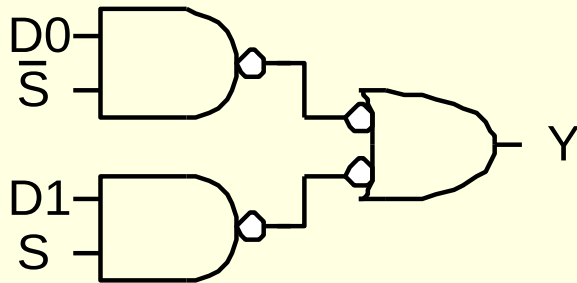
$$g_D = 8/3$$

$$g_E = 8/3$$

$$p = 16/3$$

Example

- The multiplexer has a maximum input capacitance of 16 units on each input. It must drive a load of 160 units. Estimate the delay of the NAND and compound gate designs.



$$H = 160 / 16 = 10$$

$$B = 1$$

$$N = 2$$

NAND Solution

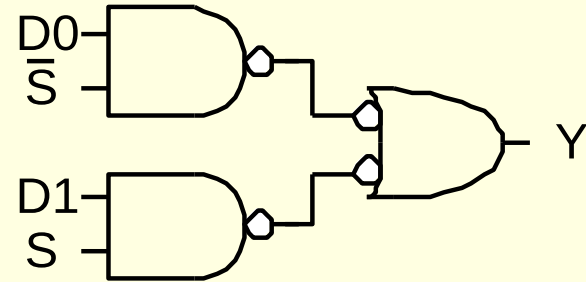
$$P = 2 + 2 = 4$$

$$G = (4/3)g(4/3) = 16/9$$

$$F = GBH = 160/9$$

$$\hat{f} = \sqrt[N]{F} = 4.2$$

$$D = N\hat{f} + P = 12.4\tau$$



$$H = 160 / 16 = 10$$

$$B = 1$$

$$N = 2$$

Compound Solution

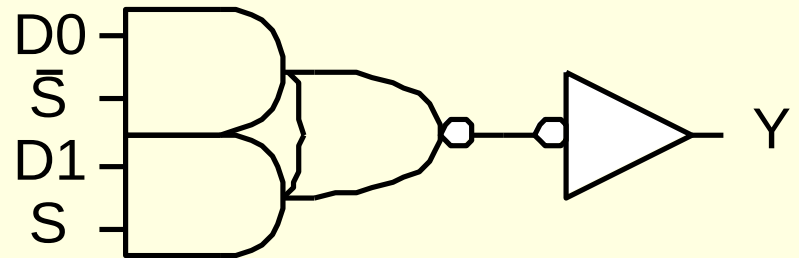
$$P = 4 + 1 = 5$$

$$G = (6 / 3)g(1) = 2$$

$$F = GBH = 20$$

$$\hat{f} = \sqrt[N]{F} = 4.5$$

$$D = N\hat{f} + P = 14\tau$$



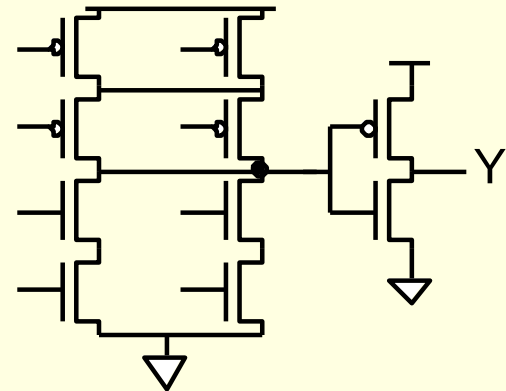
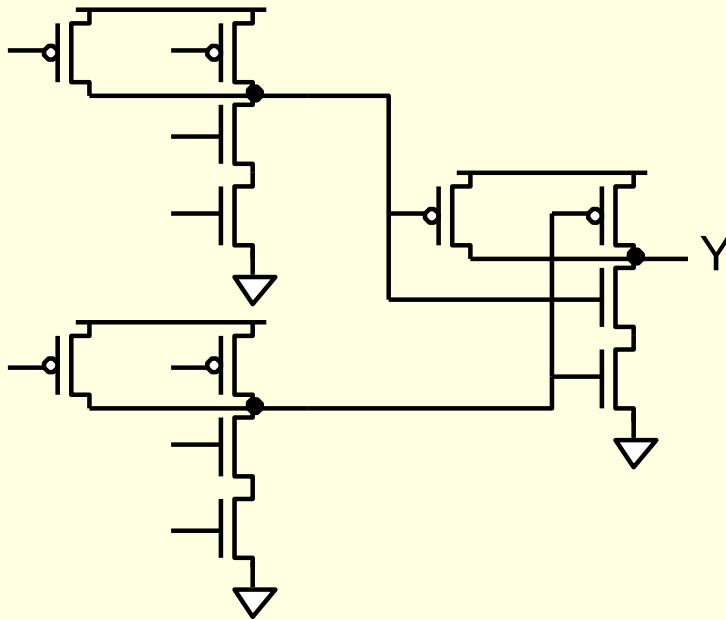
$$H = 160 / 16 = 10$$

$$B = 1$$

$$N = 2$$

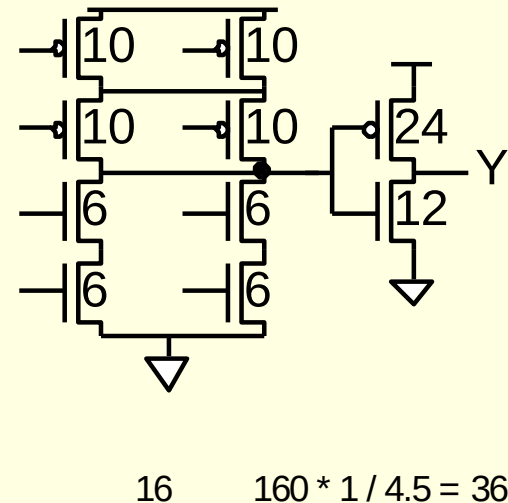
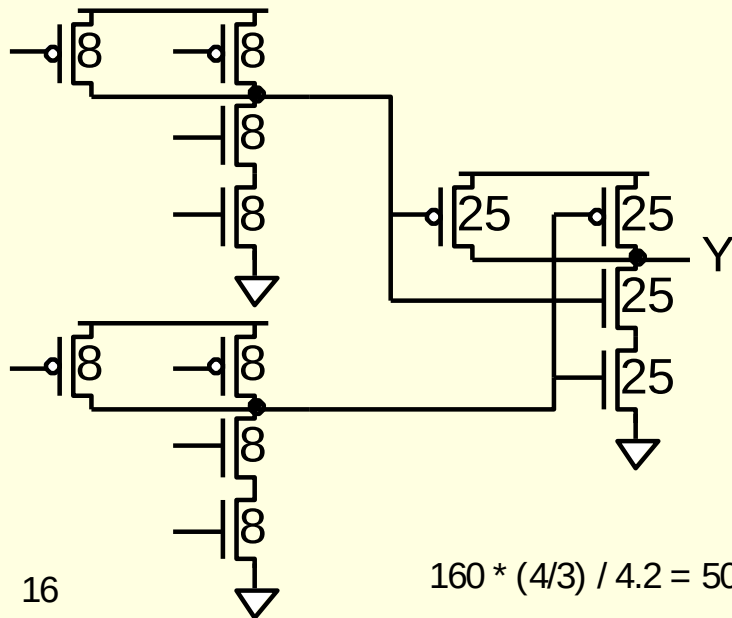
Example

- Annotate your designs with transistor sizes that achieve this delay. Each input can represent a maximum of 16λ of transistor width.



Example...Solution

- Annotate your designs with transistor sizes that achieve this delay. Each input can represent a maximum of 16λ of transistor width.



Example

- Calculate the minimum delay, in T , to compute $F = AB + CD$ using the circuits from the following figures. Each input can present a maximum of 20λ of transistor width. The output must drive a load equivalent to 100λ of transistor width. Choose transistor sizes to achieve this delay.

